

2026 Weston Calculus & Groups Review

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This is the review sheet for Calculus, Chaos Theory, and Group Theory. It will not be graded, but we will go over it in class. Some of the questions may seem familiar.

1. (a) Find the following limit: $\lim_{x \rightarrow 1} \frac{2x^2 - 2}{2x^2 + x - 3}$

$$\frac{2(x+1)\cancel{x-1}}{\cancel{x-1}(2x+3)}$$

$$\frac{4}{5}$$

$$\frac{4x}{4x+1} = \frac{4}{5}$$

- (b) For the same equation $y = \frac{2x^2 - 2}{2x^2 + x - 3}$, find a tolerance δ such that for every x between $1 - \delta$ and $1 + \delta$, y is no more than $\epsilon = 0.01$ away from the answer you found in part (a). For example, if your answer for (a) was 4.2, then your y values would have to stay between 4.19 and 4.21.

$$\delta = 0.01$$

$$\frac{3.98}{4.18}$$

$$\frac{4.02}{5.02}$$

$$\frac{1.99^2 - 2}{1.99^2 + 1.99 - 3}$$

$$\frac{2.01^2 - 2}{2.01^2 + 2.01 - 3}$$

2. Find the derivative of the function $f(x) = 6(x - 3)^2 - x + 4$

$$(x-3)^2$$

$$6 \cdot 2(x-3) \cdot 1 - 1$$

$$12(x-3) - 1$$

3. Find the derivative of the function $f(x) = \frac{e^x \sin(x)}{x^2}$

$$\frac{x^2(e^x \cos x + \sin x)e^x - e^x \sin x \cdot 2x}{x^4}$$

4. (a) Find the indefinite integral (antiderivative) $\int -\sin(x) + x^3 dx$

$$\cos x + \frac{x^4}{4} + C$$

- (b) Find the definite integral $\int_0^{\pi} -\sin(x) + x^3 dx$

$$\cos x + \frac{x^4}{4} + C \Big|_0^{\pi} \quad \left(-1 + \frac{\pi^4}{4} + C \right) - \left(1 + C \right)$$

$\frac{\pi^4}{4} - 2$

5. In words, describe what it means for a system to be chaotic

Small changes in the input
lead to big, unpredictable changes in the output

6. Give an example of a chaotic system

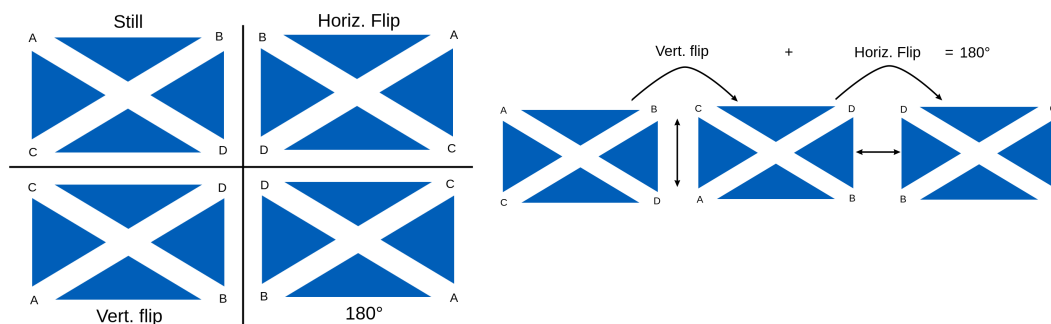
Julia set

Conway's game of life

7. Consider the flag of Scotland



There are 4 ways to move the flag that leave it appearing identical: 1. Leave it still; 2. Rotate it 180° ; 3. Flip it horizontally (left to right); 4. Flip it vertically (top to bottom). These are the flag's "symmetries".



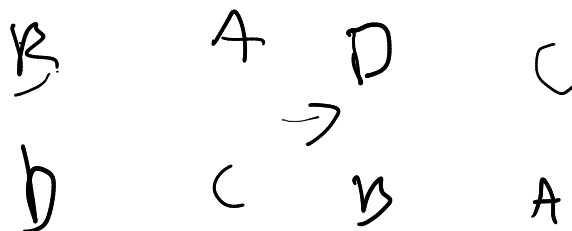
We can "compose" two symmetries by doing one and then the other, and seeing what symmetry would take the flag to that same position in a single step. For example, a vertical flip then a horizontal flip equals a 180° turn.

(a) Make a full table of all possible compositions of two symmetries of the flag.

Handwritten table of compositions of symmetries (S, V, H, R):

	S	V	H	R
S	S	V	H	R
V	V	S	R	H
H	H	R	S	V
R	R	H	V	S

8. Consider this Taiwanese recycling symbol



- (a) List all the symmetries (reflections or rotations) that leave the symbol appearing identical. (Don't forget to include leaving it still!). Moves that leave the symbol in the same position as each other should be treated as the same symmetry (e.g. a 180° clockwise turn, a 540° clockwise turn, and a 180° counterclockwise turn are all the same symmetry)

90° c 90° r 5 180° r

- (b) Make a full table of all possible compositions of two symmetries of the symbol.

2nd

	S	c	a	r
S	S	c	a	r
c	c	r	s	a
a	a	s	r	c
r	r	a	c	s

1st

$S \circ c = c$
 $S \circ c = c$
 $S \circ a = r$
 $S \circ r = a$

9. The symmetries on the shapes in the two previous problems, plus the rules for combining them, form those shapes' "symmetry groups", a common kind of "group". In general, groups consist of some basic objects ("elements") and rules for combining them (an "operation"). For symmetry groups, the elements are symmetries and the operation is combining them as described in question 5.

Two groups A and B are said to be "isomorphic" to each other if you can pair up the elements of A with those of B so that the results of the operation always match up correctly (i.e. if in A , $uv = w$ and in B , $xy = z$, and u is paired with x and v is paired with y , then w must be paired with z).

- (a) The symmetry groups of the symbol and the flag are not isomorphic to each other. Explain why no appropriate pairing between these groups can exist.

e.g. every element of the flag's group is its own inverse, but some of the symbol's are not.

- (b) Consider the group (named $\mathbb{Z}_4$) whose table is below. Its elements are the numbers 0, 1, 2, 3 and its operation is adding two numbers and taking the remainder when you divide the result by 4.

+4	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\mathbb{Z}_4$ is isomorphic to either the flag's or the recycling symbol's symmetry group, but not both. Find which one, and give one appropriate pairing between their elements.

Symbol

$0 \leftrightarrow S$
 $1 \leftrightarrow 180^\circ$
 $2 \leftrightarrow 90^\circ$
 $3 \leftrightarrow 270^\circ$

or

$0 \leftrightarrow 90^\circ$
 $1 \leftrightarrow 180^\circ$

$$\int_1^5 x^3 - 2x \, dx$$

$$\left. \frac{x^4}{4} - x^2 + c \right|_1^5$$

$$\left(\frac{5^4}{4} - 5^2 + c \right) - \left(\frac{1^4}{4} - 1^2 + c \right)$$

$$\frac{625}{4} - 25 - \left(\frac{1}{4} - 1 \right)$$

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$$\int e^{(2x+1)} \, dx$$

$$u = 2x+1$$

$$du = 2 \, dx$$

$$\frac{1}{2} \int 2 e^{(2x+1)} \, dx$$

$$u = 2x+1$$

$$\frac{du}{dx} = 2 \Rightarrow du = 2 \, dx$$

$$\frac{1}{2} \int e^u \, du$$

$$\frac{1}{2} e^u = \frac{1}{2} e^{2x+1}$$

$$\dot{x} = \frac{dx}{dt} - kx$$

$$\ddot{x} = \frac{d^2x}{dt^2} \quad \ddot{x} = -\frac{kx}{m}$$

$$\sin y + y^3 = 6 - x^3$$

$$\cos(y) \cdot \frac{dy}{dx} = -3x^2$$

$$+ 3y^2 \cdot \frac{dy}{dx} \quad \frac{dy}{dx} = \frac{-3x^2}{\cos y + 3y^2}$$

$$0. \overline{99} = 1$$

$$\cdot 10$$

$$\cdot 10$$

$$0. \overline{99} = x$$

$$9. \overline{99} = 10x$$

$$9. \overline{99} = 10$$

$$- 9$$

$$- 9$$

$$0. \overline{99} = 10x - 9$$

$$x = 10x - 9$$

$$- 9x = -9$$

$$x = 1$$

$$. \overline{99} = 1$$